Pattern Recognition / NYCU Fall 2026 Assignment #1 (due: 10/16/2026)

The goal of this assignment is to let you experiment with linear feature spaces. Use **python with numpy** for your experiments. You are allowed to use existing libraries/modules/toolboxes or AI generated codes. However, you need to understand how they work. You are responsible to make your program organized and correct.

Your program/report should include the following:

1. Synthetic data specification:

1. **Classes ($C = 3$):** Generate $N = 300$ samples total (100 samples per class).
2. **True Signal Subspace ($d_{\text{signal}} = 3$):**
   - Class 1 mean: $\boldsymbol{\mu}_1 = [+2, +2, 0]^T$
   - Class 2 mean: $\boldsymbol{\mu}_2 = [-2, +2, 0]^T$
   - Class 3 mean: $\boldsymbol{\mu}_3 = [0, -2, 0]^T$
   - Shared within-class covariance in signal dimensions: $\Sigma_{\text{signal}} = \begin{bmatrix} 1.0 & 0.6 & 0.0 \\ 0.6 & 1.5 & 0.0 \\ 0.0 & 0.0 & 0.5 \end{bmatrix}$.
3. **Noise Subspace (d_{noise} variable):**
   - Append d_{noise} uninformative noise dimensions generated from $\mathcal{N}(0, \sigma^2_{\text{noise}})$, where $\sigma^2_{\text{noise}} = 0.5$.
   - Total feature dimension: $d = d_{\text{signal}} + d_{\text{noise}} = 3 + d_{\text{noise}}$.

   Make plots (projected onto 2-D) of the dataset with equal-Mahalanobis-distance contours. See how they look like in different views (projections).

   Also make 3-D scatter plots with only the initial 3 signal dimensions together with 95%-confidence ellipsoids for each class.

2. Empirical observation of the Hughes phenomenon:

   Classifiers to use: (1) 1-NN; (2) Mahalanobis distance based.

   Split data into 50/50 training and testing subsets.

   Vary the noise dimension (from 1 to 100) and plot the curves of training and testing accuracies.

   Vary the splitting (e.g., 20/80 training/testing) to observe whether there are changes.

3. Spatial whitening with PCA and ridge regularization:

   Plot the eigenvalue spectrum (eigenvalues in descending order).

   Repeat Part 2 by applying PCA first to the dataset (with 100 noise dimensions). Vary the projected dimensionality from 1 to 100 (using highest-eigenvalue axes) and plot the curves of training and testing accuracies.

   Include ridge regularization in PCA (with varying γ) and compare their testing accuracy plots.

4. Supervised dimensionality reduction:

   Apply LDA to the data with 100 noise dimensions. Compare with PCA results.

   Compare LDA and regularized LDA results.

5. Iris dataset experiment:

   Get the Iris dataset. Repeat experiments 2-4 (with your own added 100 noise dimensions). Focus on the Mahalanobis distance based classifier only.

Compare results with and without data normalization of the four features of the Iris dataset.

Observe how the classification accuracy changes with different levels of noise variance.

Write a report (limited to 5 pages, single-spaced) describing the following, according to the requirements mentioned above:

- Methods you have implemented.
- Experiments you have done, and the results. You can include additional experiments not stated above.
- Analysis - Are the results what you expect? Why?

The report (PDF format only) is to be submitted electronically through e3. Also include the program listing (in *.py or *.jpynb formats) as an appendix to your report (in the same file, not counted toward the page limit).

You can write your report in English or Chinese.

Late submission policy: 10% credit deduction for each day late; up to 3 days late accepted.